$$= 2000 + \left(\frac{100 - 74}{54}\right) \times 1000$$
$$= 2000 + \left(\frac{26}{54}\right) \times 1000 = 2000 + 481.5$$
$$= 2481.5$$

Example 8.13 The Median of the following data is 24. Find the value of x.

Class Interval (CI)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Frequency (f)	6	24	x	16	9

Solution

Class Interval (CI)	Frequency (f)	Cumulative frequency (cf)
0-10	6	6
10-20	24	30
20-30	x	$30 + x$
30-40	16	$46 + x$
40-50	9	$55 + x$
	$N = 55 + x$	

Since the median is 24 and median class is 20 – 30

$l = 20 \quad N = 55 + x, \quad m = 30, \quad c = 10, \quad f = x$

$$\text{Median} = l + \frac{\left(\frac{N}{2} - m\right)}{f} \times c$$

$$24 = 20 + \frac{\left(\frac{55 + x}{2} - 30\right)}{x} \times 10$$

$$4 = \frac{5x - 25}{x} \quad \text{(after simplification)}$$

$$4x = 5x - 25$$

$$5x - 4x = 25$$

$$x = 25$$

Note

The median is a good measure of the average value when the data include extremely high or low values, because these have little influence on the outcome.

Exercise 8.2

1. Find the median of the given values : 47, 53, 62, 71, 83, 21, 43, 47, 41.
2. Find the Median of the given data: 36, 44, 86, 31, 37, 44, 86, 35, 60, 51

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